Cyclistic Bike-Share Analysis

How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?

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Project: Google Data Analytics Capstone Case Study
Tools: Google Colab, Python, and data visualization tools
Dataset period: October 2025 – September 2026 *(confirm against your actual dataset)*

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1. Ask — Define the Business Task

Business Background

Cyclistic is a fictional bike-share company operating in Chicago. Its service allows customers to unlock bicycles at one station and return them to another.

Cyclistic has two main customer groups:

- **Casual riders:** Customers who purchase single-ride or full-day passes.
- **Annual members:** Customers who purchase annual memberships.

The company's finance analysts have determined that annual members are more profitable than casual riders. Therefore, Cyclistic wants to understand how these customer groups use its bicycles differently.

Business Task

The goal of this analysis is to identify differences in bike usage between casual riders and annual members. These insights can help Cyclistic develop marketing strategies to encourage casual riders to purchase annual memberships.

Key Business Question

How do annual members and casual riders use Cyclistic bikes differently?

Stakeholders

- Cyclistic marketing analytics team
- Cyclistic executive team
- Marketing decision-makers responsible for membership growth

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2. Prepare — Understand the Data

Data Source

This project uses Cyclistic historical bike-trip data made available for the Google Data Analytics case study. The original case study identifies Motivate International Inc. as the data provider and notes that the data is subject to its applicable license.

Data Period

The analysis covers **October 2025 through September 2026**, subject to verification against the files actually analyzed.

Data Organization

The trip dataset contains information used to examine bike rides, including trip start and end times, rider type, and other available trip attributes.

Data Credibility and Limitations

The analysis uses historical trip records to compare riding behavior. The dataset does not provide permission to identify individual riders or link their rides to personal payment information.

The findings describe observed usage patterns. They do not, by themselves, establish why a casual rider would purchase an annual membership.

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3. Process — Clean and Prepare the Data

Tools Used

Python in Google Colab was used to combine the monthly trip data files, inspect the dataset, clean the records, calculate ride durations, and prepare the data for analysis.

Data Cleaning Steps

The following steps were performed to prepare the dataset:

1. Imported and combined the 12 monthly CSV files into one dataset.
2. Inspected the dataset structure and relevant columns.
3. Converted date and time fields to datetime format where needed.
4. Calculated ride duration in minutes using the ride start and end times.
5. Filtered the data to include the analysis period from October 1, 2025, through September 30, 2026.
6. Excluded rides with durations of zero minutes or less and rides longer than 1,440 minutes (24 hours).
7. Prepared the cleaned data for analysis by rider type, ride duration, day of the week, and month.

Data Quality Checks

The final analysis dataset contained **6,166,346 rides**. The checks performed on the cleaned dataset found:

- Missing ride IDs: 0
- Duplicate ride IDs: 0
- Invalid ride durations: 0

These checks supported the use of the cleaned dataset for the analysis. The results reported here describe the final dataset; the number of records excluded during cleaning was not documented in the results summarized for this report.

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4. Analyze — Key Findings

4.1 Overall Ride Volume by Rider Type

The analysis included 6,166,346 rides recorded between October 2025 and September 2026.

- **Annual members:** 3,981,574 rides (64.5%)
- **Casual riders:** 2,184,772 rides (35.5%)

Annual members accounted for approximately 64.5% of all rides, while casual riders accounted for 35.5%. This indicates that annual members generated the larger share of recorded trips during the study period.

Business insight: Cyclistic should continue supporting its annual membership base while investigating opportunities to convert casual riders into members.

4.2 Average Ride Duration

The analysis compared the mean and median ride durations for both rider groups.

Metric	Casual riders	Annual members
Number of rides	2,184,772	3,981,574

Average duration	17.56 minutes	11.92 minutes
Median duration	10.75 minutes	8.51 minutes

Casual riders averaged 17.56 minutes per ride, compared with 11.92 minutes for annual members. Casual rides were approximately 47.3% longer on average.

The median ride duration was also higher for casual riders, at 10.75 minutes compared with 8.51 minutes for members.

Business insight: Casual riders tend to take longer rides, while annual members generate more trips. Cyclistic could investigate whether recreational riding, commuting, or other trip purposes help explain these differences.

4.3 Weekday Versus Weekend Usage

The distribution of rides within each customer group showed different patterns.

Day type	Casual riders	Annual members
Weekdays	62.70%	76.73%
Weekends	37.30%	23.27%

Approximately 37.3% of casual rides occurred on weekends, compared with 23.27% of member rides. Annual members had a larger share of weekday rides.

Business insight: Casual riders have a relatively stronger weekend usage pattern. Cyclistic could test weekend-focused membership promotions while continuing to explore the needs of frequent weekday riders.

4.4 Rides by Day of the Week

The busiest day differed between the two customer groups.

- **Casual riders:** Saturday, with 464,665 rides.
- **Annual members:** Wednesday, with 644,595 rides.

Casual rider activity peaked on Saturday, while member activity peaked on Wednesday. Saturday was also the busiest day for casual riders by a substantial margin compared with their weekday counts.

Business insight: Saturday may be a useful time to test membership messaging aimed at casual riders. The Wednesday peak among members suggests that weekday usage should also be considered when studying member behavior.

4.5 Monthly Ride Trends

Monthly ride counts showed considerable variation across the year.

- Total rides were highest in July 2026, at 868,326.
- August 2026 recorded 867,421 rides.
- January 2026 recorded the lowest total, at 137,589.
- Casual rides increased from 24,656 in January to 357,121 in July.
- Member rides increased from 112,933 in January to 520,948 in August.

Both rider groups recorded substantially more rides during the warmer months than in winter. Casual rides also showed a particularly strong summer increase.

Business insight: Cyclistic could test seasonal marketing campaigns before and during periods of increasing ride activity, particularly in late spring and summer. The effectiveness of such campaigns should be measured rather than assumed.

4.6 Overall Analysis Summary

The results reveal four key differences between casual riders and annual members:

1. Annual members account for the majority of recorded rides.
2. Casual riders have longer average and median ride durations.
3. Casual riders have a higher proportion of weekend rides, while annual members have a higher proportion of weekday rides.
4. Ride activity changes throughout the year, with both groups recording their highest monthly totals during summer.

Together, these patterns provide a basis for testing targeted membership marketing strategies. However, trip data alone cannot establish why an individual chooses casual riding instead of an annual membership. Additional customer research and campaign testing would help explain these behaviors.

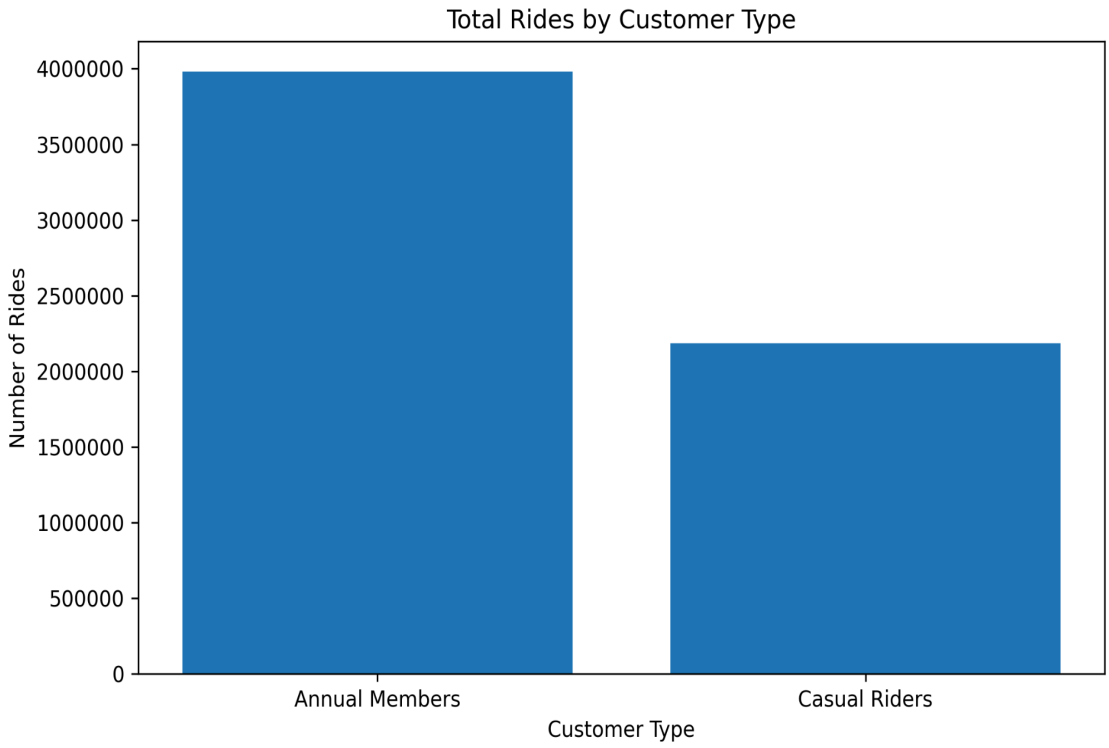
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5. Share — Visualizations

The following visualizations show how Cyclistic annual members and casual riders use bikes differently. The charts compare ride volume, average ride duration, day-of-week patterns, and monthly trends from October 2025 through September 2026.

Visualization 1: Weekday vs. Weekend Rides by Customer Type

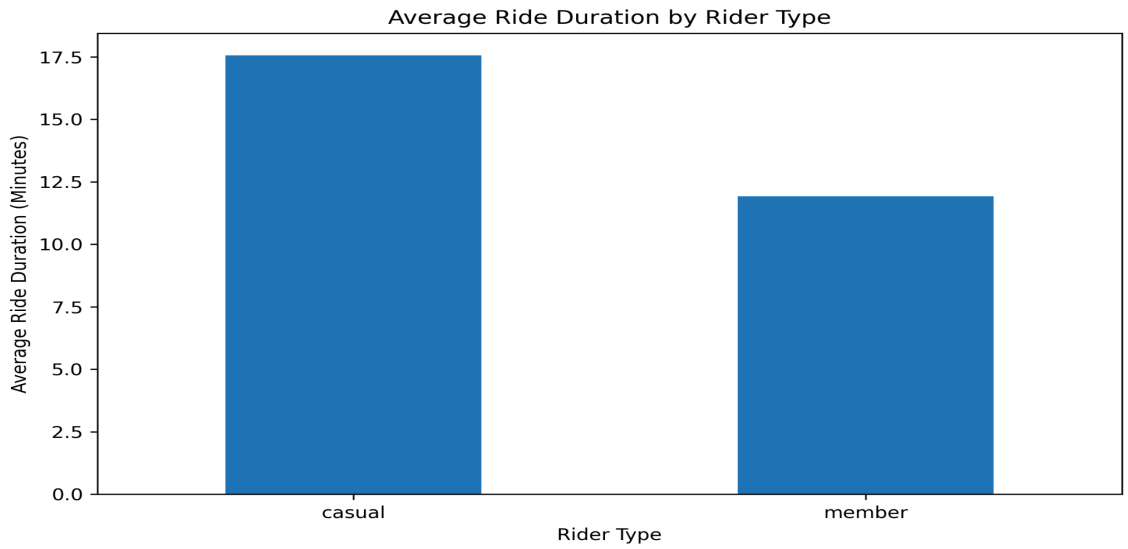
Chart title: Total Rides by Customer Type



Key Takeaway: Members took 76.73% of their rides on weekdays and 23.27% on weekends. Casual riders had a more balanced pattern, with 62.7% of rides occurring on weekdays and 37.3% on weekends. This suggests that members may rely more on Cyclistic for weekday travel, while casual riders account for a larger share of weekend activity proportionally.

Visualization 2: Average Ride Duration by Customer Type

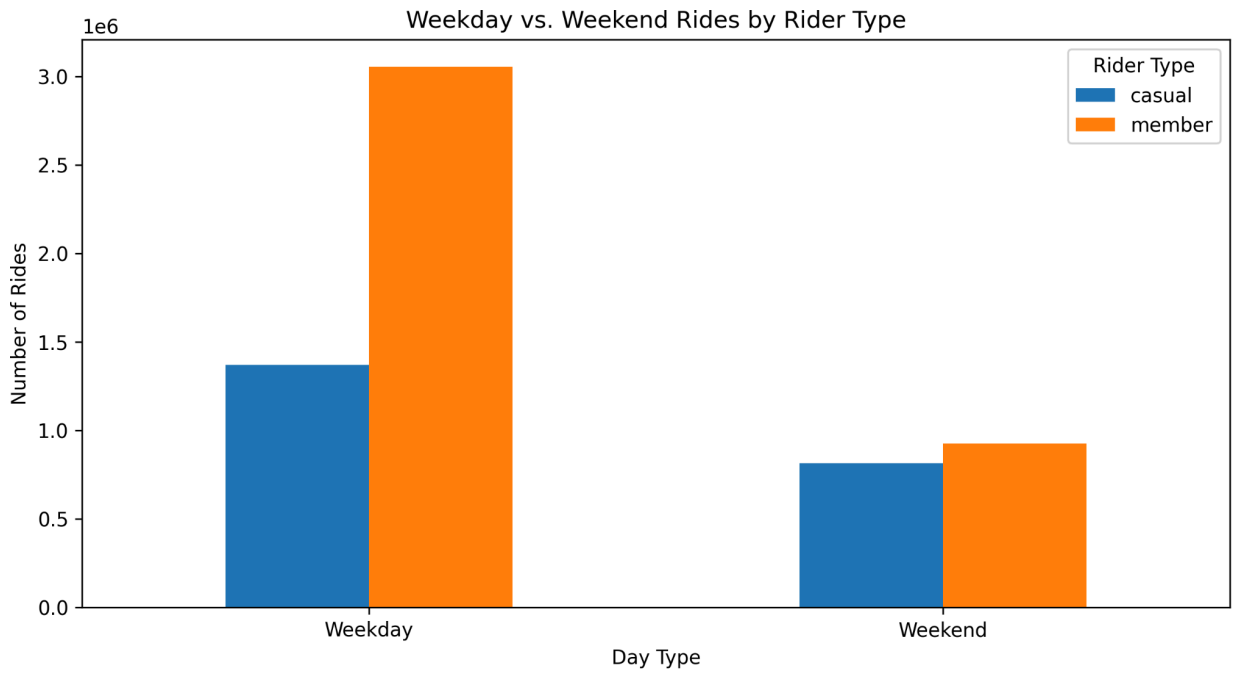
Chart title: Average Ride Duration: Members vs. Casual Riders



Key takeaway: Casual riders took longer rides on average than annual members. The average ride duration was **17.56 minutes for casual riders**, compared with **11.92 minutes for members**. Casual riders' average rides were approximately **47% longer**. This suggests that casual riders may use the bikes for leisure, sightseeing, or occasional trips, while members may use them more frequently for shorter, routine journeys. These are possible explanations that would require additional analysis to confirm.

Visualization 3: Rides by Day of Week

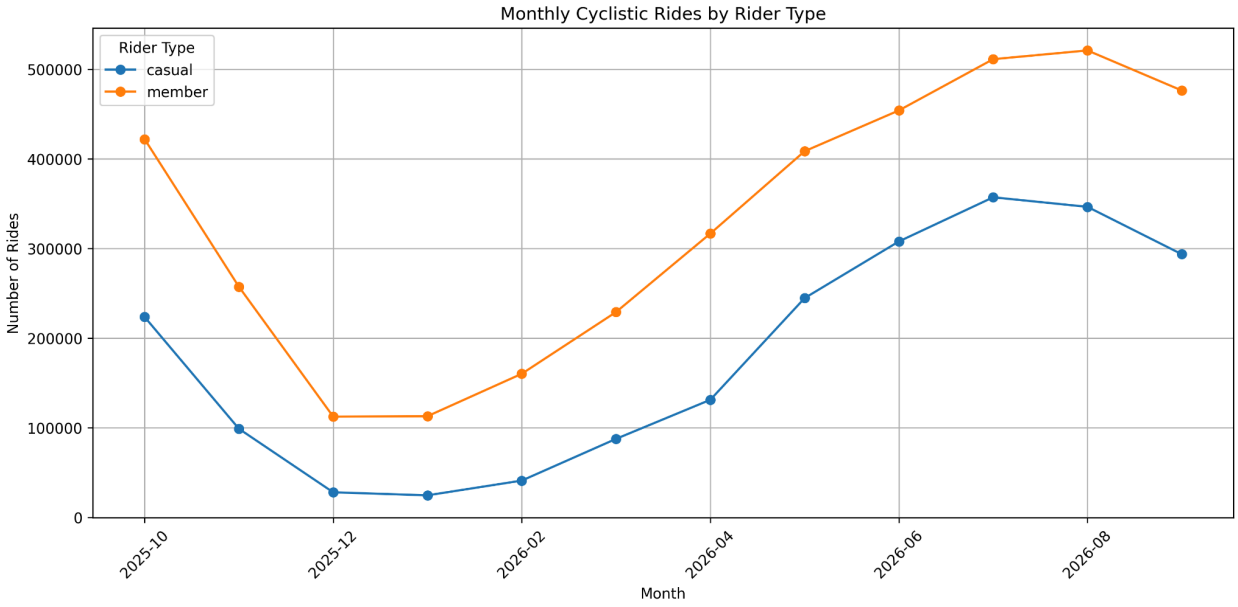
Chart title: Total Rides by Day of Week and Customer Type



Key takeaway: Members showed a stronger weekday usage pattern, while casual riders were more active on weekends. Approximately **76.73% of member rides occurred on weekdays**, compared with **62.7% of casual rides**. Saturday was the busiest day for casual riders, with **464,665 rides**, while Wednesday was the busiest day for members, with **644,595 rides**. This difference suggests that weekend-focused promotions may be a useful way to encourage casual riders to consider annual memberships.

Visualization 4: Monthly Ride Trends

Chart title: Monthly Rides by Customer Type (October 2025–September 2026)



Key takeaway: Ride volume changed substantially throughout the year, with the highest total in **July 2026 at 868,326 rides**, followed closely by August at **867,421 rides**. January had the lowest total, with **137,589 rides**. Both customer groups experienced higher ride volumes during the warmer months and lower volumes during winter. Casual rides increased from **24,656 in January** to **357,121 in July**, while member rides peaked at **520,948 in August**. These seasonal patterns suggest that Cyclistic could test targeted promotions before and during the spring and summer riding seasons.

Overall Visualization Summary

Across all four visualizations, annual members generated more rides overall and showed a stronger weekday usage pattern. Casual riders took longer rides and had a greater concentration of weekend activity. Both groups showed clear seasonal changes in ride volume. These findings support testing weekend-focused membership promotions and seasonal campaigns to encourage casual riders to become annual members.

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6. Act — Recommendations

The final recommendations will be based on the verified analysis results.

Recommendation 1: Target Marketing to Casual Riders

Use the identified riding patterns to develop targeted membership campaigns for casual riders during the periods when they are most active.

Recommendation 2: Highlight the Value of Annual Membership

Use Cyclistic's membership pricing and benefits to explain why an annual membership may be valuable to frequent casual riders. Validate the value proposition with actual pricing and customer research.

Recommendation 3: Test Data-Driven Campaigns

Run targeted marketing experiments based on observed riding patterns. Compare membership conversion rates across campaign groups to determine which messages and timing are most effective.

These recommendations are initial hypotheses. They should be refined after reviewing the actual findings and any additional customer or marketing data.

Conclusion

This analysis examined Cyclistic bike-share usage patterns among annual members and casual riders from October 2025 through September 2026. The final dataset contained 6,166,346 rides.

The results showed that annual members accounted for approximately 64.6% of rides, while casual riders accounted for 35.4%. Casual riders had a longer average ride duration than members (17.56 minutes compared with 11.92 minutes). Members also completed a larger share of their rides on weekdays, while casual riders had a higher proportional share of weekend rides.

These findings suggest opportunities to explore converting casual riders into annual members through targeted marketing, membership-value messaging, and campaigns informed by when casual riders are most active. Any proposed campaign should be tested and evaluated using measurable conversion results.

This case study demonstrates how data cleaning, analysis, and visualization can reveal differences in customer behavior and support data-informed business recommendations.

Project Deliverables

- Business task and objectives
- Data source description
- Data cleaning documentation
- Analysis summary with verified figures
- Charts with findings
- Three evidence-based recommendations
- Final project files and portfolio description